

# Model Card + Data Card

Articulate AI · CUET AI Hackathon 2026

## MODEL CARD

### Azure Cognitive Services — Pronunciation Assessment

REST API · southeastasia

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| Input        | Raw audio (webm / mp4 / ogg) + reference text                          |
| Output       | overall_score, accuracy, fluency, completeness, per-phoneme IPA scores |
| Granularity  | Phoneme-level IPA · EnableMiscue: true                                 |
| Voices (TTS) | JennyNeural (F) · GuyNeural (M) · AndrewNeural (M)                     |
| Use case     | Word pronunciation test · Sentence test · IELTS turns (two-pass)       |
| Limitation   | Requires quiet environment · Best with 16kHz mono WAV                  |

### OpenAI GPT-4o-mini

temperature 0.3–0.9 · max 150 tokens

|               |                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------|
| Input         | Prompt + phoneme scores / transcript / chat history                                               |
| Output        | JSON or plain text — Bangla tips, band reports, grammar corrections                               |
| Tasks         | IELTS examiner · band report · Bangla phoneme tip · grammar check · word lookup · exam generation |
| Bangla output | Yes — all tips, reports, error messages in Bengali (বাংলা)                                        |
| Limitation    | Hallucination risk on phoneme specifics · capped at 150 tokens for chat                           |

### DeepFilterNet3 — Neural Noise Suppressor

Python FastAPI · PyTorch · 48kHz

|              |                                                                     |
|--------------|---------------------------------------------------------------------|
| Input        | Raw browser audio (any format, ffmpeg decodes)                      |
| Output       | Denoised 16kHz mono WAV — ready for Azure                           |
| Architecture | DeepFilterNet3 trained on 10K+ hrs of diverse noise                 |
| Noise types  | Non-stationary: traffic, babble, wind — not just hum                |
| Post-process | Peak-normalized to match input level (prevents 25–40dB volume drop) |
| Fallback     | Gracefully skips to raw audio if microservice is offline            |

## DATA CARD

### Oxford 3000 Vocabulary Corpus

Oxford University Press · CEFR A1–C1

7,640  
Words

160  
Lesson Phrases

7,800+  
Audio Files

|          |                                                                                     |
|----------|-------------------------------------------------------------------------------------|
| Source   | Oxford 3000 PDF — custom 4-column-aware parser                                      |
| Schema   | word, IPA, bangla_meaning, syllables, CEFR level, frequency rank, part_of_speech    |
| Audio    | Male (AndrewNeural) + Female (JennyNeural) · Azure TTS · 24kHz MP3 · Cloudinary CDN |
| Phrases  | GPT-4o-mini generated per-lesson using that lesson's exact vocabulary               |
| Language | English (source) · Bengali/Bangla (meaning, feedback)                               |
| Storage  | PostgreSQL on Neon · Audio on Cloudinary · 7-day localStorage cache                 |
| License  | Oxford 3000 word list — educational use                                             |

### User Pronunciation & Progress Data

PostgreSQL · Neon Serverless · 25+ tables

SM-2  
SRS Algorithm

4 States  
NEW → MASTERED

10 Attempts  
Sparkline History

|               |                                                                                |
|---------------|--------------------------------------------------------------------------------|
| Collected     | Per-word pronunciation scores (0–100%), phoneme breakdowns, attempt timestamps |
| SRS fields    | interval, ease_factor, streak, familiarity (NEW/LEARNING/FAMILIAR/MASTERED)    |
| XP log        | user_xp_log — reason, amount, timestamp for every XP event                     |
| Risk phonemes | /v/ /æ/ /θ/ /ð/ /z/ /ɜ/ /w/ — flagged per session, stored per user             |
| Privacy       | Audio not stored — only scores + transcripts · JWT-authenticated access        |
| Retention     | Sparkline: last 10 attempts per word (localStorage) · Full history in Neon DB  |